

UPSC Prelims

Environment & Ecology

Class No. 3

Biodiversity and Indian Laws

Biodiversity and Conservation



- Broad Topics to be covered-
- Meaning of Biodiversity
- Importance
- Biodiversity of India and World
- Threats
- Conservation measures- in situ and ex situ

Meaning and levels of Biodiversity



- 1992 UN Earth Summit (Rio Summit) define biodiversity as
- “Variability among living organisms from all sources, including terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part. This includes diversity within species, between species and ecosystems of a region.”
- Introduced by Walter Rosen (1986)
- Edward Wilson popularized the term ‘Biodiversity’ to describe diversity at all levels of biological organization from populations to biomes.

Meaning and levels of Biodiversity



- 3 levels of biodiversity
- **Genetic**- differences in genetic make-up (number and types of genes) between distinct species and to the genetic variation within a single species
 - Continental tiger (*Panthera tigris tigris*)
 - >Bengal
 - >Indochinese
 - >Malayan
 - >Amur (Siberian) tiger

Meaning and levels of Biodiversity



- **Species-** diversity refers to the variety in number and richness of the species in any habitat
 - Continental tiger (*Panthera tigris tigris*)
 - Bengal, Malayan, Indochinese and Amur (Siberian) tiger
 - Sunda tiger (*Panthera tigris sondaica*)
 - Sumatran tiger (Java and Bali- extinct)
- **Ecosystem-** Tropical rainforest vs tropical grasslands

Meaning and levels of Biodiversity

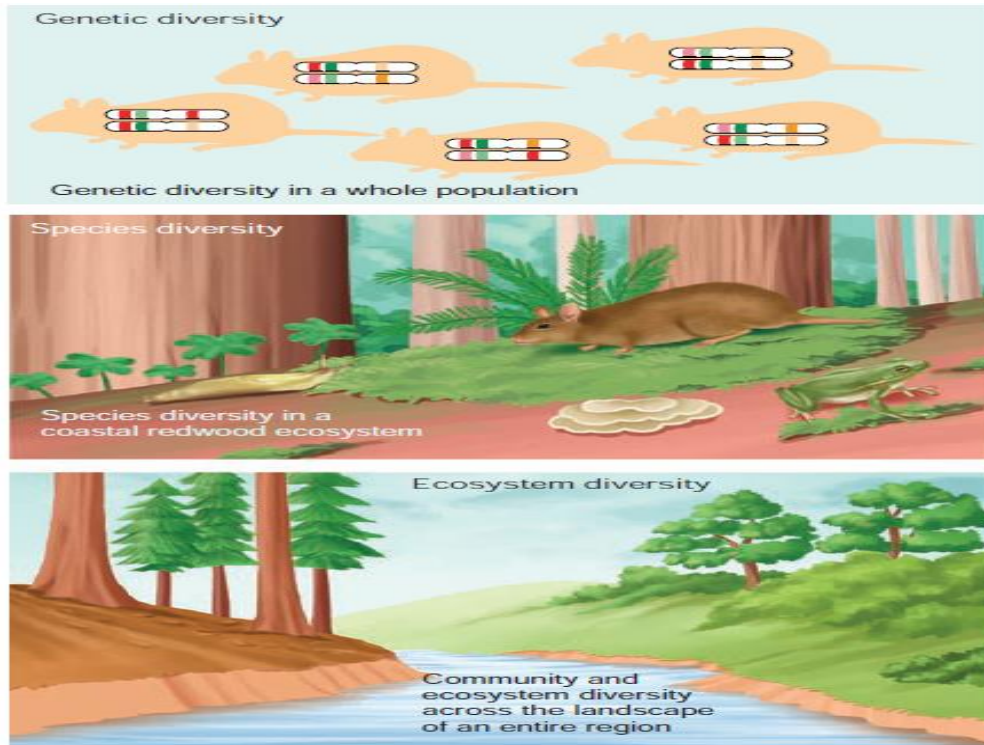


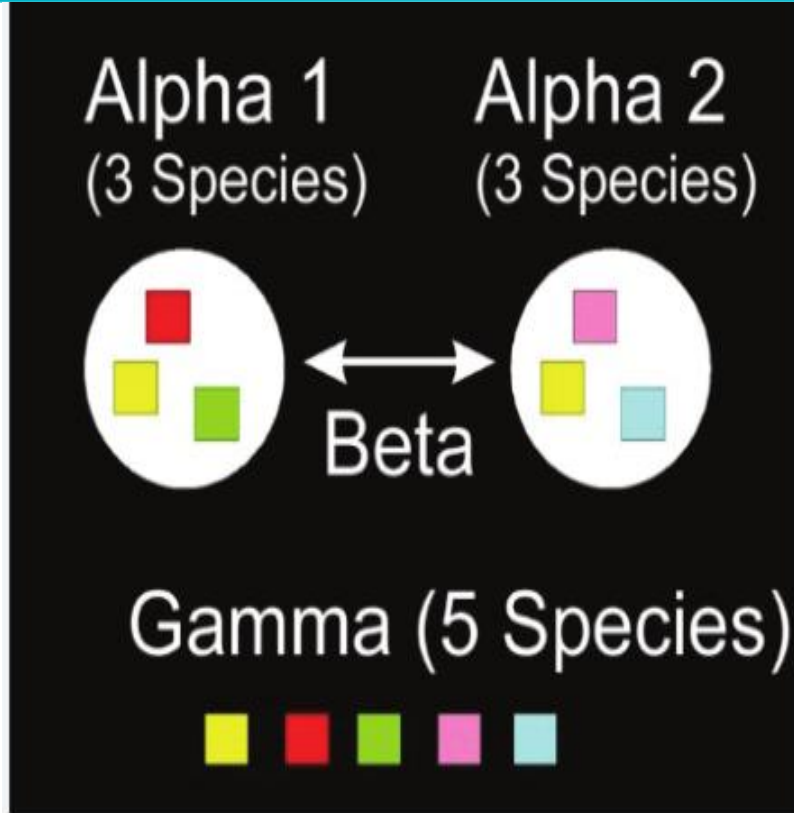
Fig. 12.1 Three levels of biodiversity

Measurement of Biodiversity



- Indices of diversity
 - Alpha- within community diversity : diversity of organisms sharing the same community / habitat
 - Beta- between community diversity : diversity of organisms sharing 2 habitat
 - Gamma- diversity over total landscape

Measurement of Biodiversity



Measurement of Biodiversity



- Species richness vs species evenness
 - Richness- different kinds of species found in a habitat
 - Evenness- distribution of species in a habitat

- Richness= same
- Evenness more in Habitat 1

Species	Habitat 1	Habitat 2
Tiger	35	5
Elephant	35	10
Chital	30	85
Total	100	100

- Few indices- Shannon-Weiner Species Diversity Index, Simpson's Diversity Index

Biodiversity hotspots



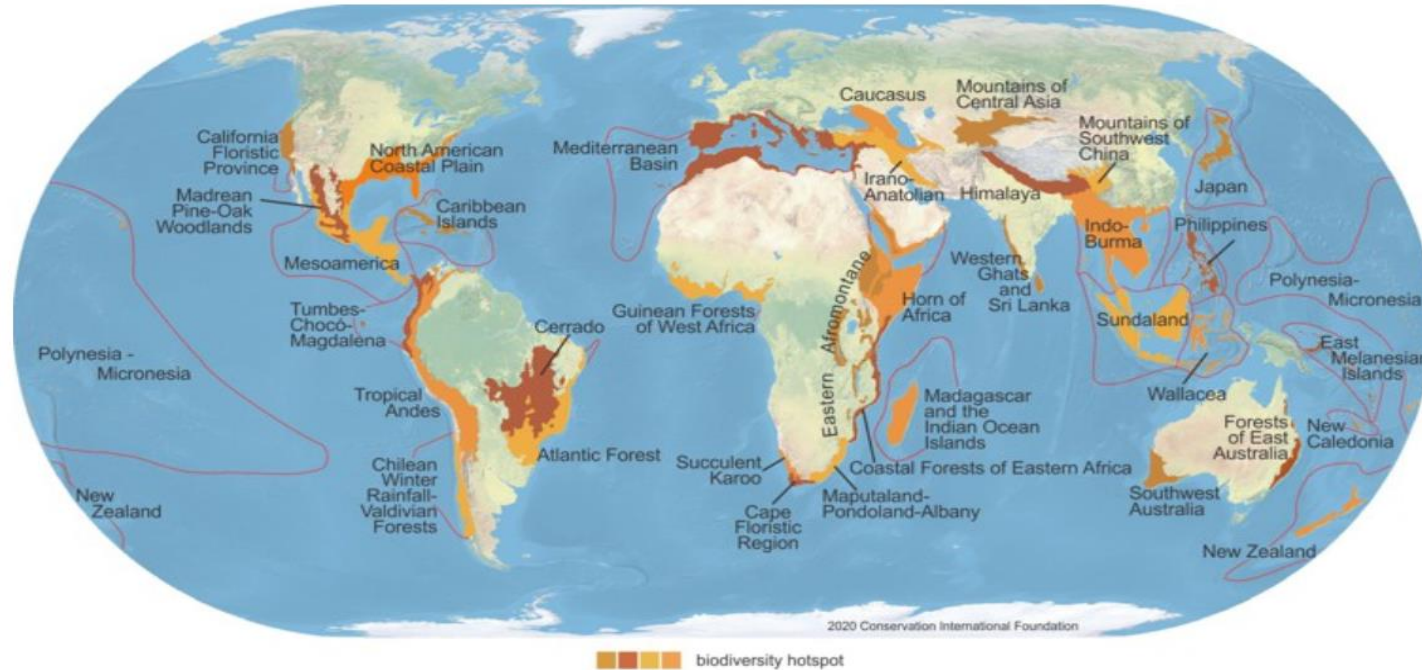
- The concept of biodiversity hotspots was given by Norman Myers
- Conservation International (CI) adopted Myers' hotspots as its institutional blueprint in 1989, and in 1996
 - CI- American nonprofit environmental organization founded in 1987
- 36 regions where success in conserving species can have an enormous impact in securing our global biodiversity.
- The forests and other remnant habitats in hotspots represent just 2.5% of Earth's land surface but 43% of bird, mammal, reptile and amphibian species as endemics.

Biodiversity hotspots



- Criterion (strict)
 - Contain at least 1,500 species of vascular plants found nowhere else on Earth (known as "endemic" species).
 - Have lost at least 70 percent of its primary native vegetation due to human action.

Biodiversity hotspots



Biodiversity hotspots



- Biodiversity Hotspots of India
 - Western Ghats
 - North east (except Assam) and Andaman (part of Indo-Burma hotspot)
 - Nicobar (part of Sundaland hotspot)
 - Himalaya

Biodiversity hotspots



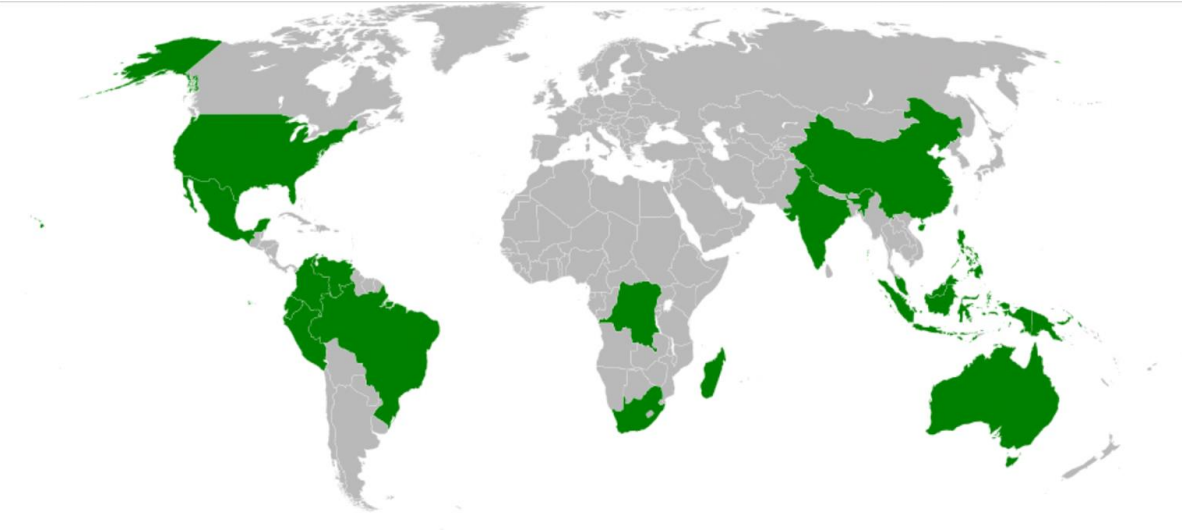
- Hottest of hot spots - 8
 - Indo – Burma region
 - Sundaland
 - Philippines
 - Western Ghats & Srilanka
 - Madagascar
 - Eastern arc & coastal forests of Tanzania/Kenya
 - Caribbean island
 - Brazil's Atlantic forests

Biodiversity hotspots



- Megadiverse countries
 - 1988, by UNEP
 - 17 countries
 - Criteria
 - Have at least 5000 of the world's plants as endemics
 - Must have marine ecosystems.

Biodiversity hotspots



-  Australia
-  Brazil
-  China
-  Colombia
-  Democratic Republic of the Congo
-  Ecuador
-  India
-  Indonesia
-  Madagascar
-  Malaysia
-  Mexico
-  Papua New Guinea
-  Peru
-  Philippines
-  South Africa
-  United States
-  Venezuela

Biodiversity hotspots



- India
 - 4 biodiversity hotspots
 - 3 hottest of hotspots
 - 1/17 megadiverse countries

Like Minded Megadiverse Countries



- Concept originated in the Cancun summit in 2002
- Only criteria is that country should be diverse and have associated traditional knowledge.
- Therefore, a landlocked country can be LMMC but not megadiverse country.
- Ex : India is Both LMMC and Megadiverse.
- USA is Megadiverse but not LMMC.
- Kenya is LMMC but not Megadiverse.

Importance of Biodiversity



- Economic value
 - Food, fuel, fodder, medicines, breeding stocks
- Ecosystem services
 - Soil formation, carbon sequestration, water resources, nutrient recycling
- Aesthetic value
 - Ecotourism, literature (Daffodils)
- Spiritual, cultural, religious value
 - Onam, Pongal
 - Animism. Eg- Kantara movie
 - Sacred groves

Importance of Biodiversity



- Knowledge value
 - Eg- traditional medicines
 - Traditional farming
- Ethical value
 - Movements- Chipko, Appiko, Silent Valley, Save Aarey

- Globally Important Agricultural Heritage Systems (GIAHS)
 - Koraput Traditional Agriculture, Odisha
 - Kuttanad Below Sea Level Farming System, Kerala
 - Pampore Saffron Heritage, Jammu & Kashmir

Threats to Biodiversity



- Identified by Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES)
 - Founded in 2012
 - India is a member
 - IPCC equivalent for Biodiversity
 - IPCC = Intergovernmental Panel on Climate Change

Threats to Biodiversity



- Land Use change
 - Deforestation, mining, fragmentation of habitats
- Climate change
 - IUCN- if 2°C rise in the earth's temperature : 6th mass extinction
- Overexploitation
 - Hunting, poaching

Threats to Biodiversity



- Invasive alien species
 - Out-compete local and indigenous species for natural resources
 - Multiply quickly due to no competitors / predators.
 - Start to Dominate ecosystems.
 - High dispersal rate.

Threats to Biodiversity



- Invasive alien species
 - Prosopis juliflora : madras HC asked TN government to get rid of this invasive weed.
 - Lantana Camara : american weed, introduced by british in india.
 - House gecko : Lizard
 - Tilapia fish : african cat fish, brought for aquaculture, now banned.
- These species were viable due to fast breeding.
- Burmese python : in the everglades in florida.

Threats to Biodiversity



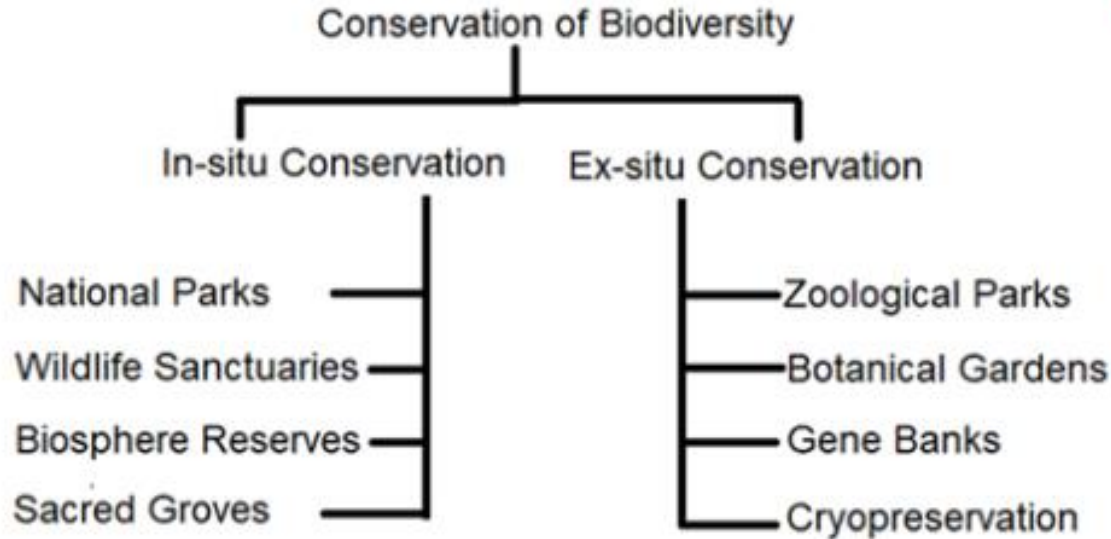
- Burmese python was introduced in the everglades in florida to control the rodent population. However, they became an invasive alien species. Now florida has hired irular tribesmen to catch these snakes.
- Irular : PVTG.

Threats to Biodiversity



- Pollution
 - Great Pacific garbage
 - Ganga- gharial, dolphin, turtle
- Genetic pollution
 - GM Crops
 - GM Mustard (DMH 11)- limited trials
 - GM crops carry genes that would pollute the native gene pool and might lead to elimination of natural varieties.

Biodiversity Conservation



Biodiversity Conservation



- Ex Situ : outside natural habitat.
 - Herbarium (plants), Arboretum (trees)
 - Botanical garden
 - Acharya Jagadish Chandra Bose Indian Botanic Garden in Howrah- India's largest

Biodiversity Conservation



Seed Bank : Places where seeds are stored at ultra low temperature

Known as cryopreservation

World's largest seedbank is at svalbard, Norway. It is a doomsday bank.

India has contributed to svalbard bank.

Changla, Ladakh : India's seedbank

Biodiversity Conservation



- Captive Preservation
 - Vulture breeding sites : jatayu breeding center
 - Gharial : Chambal river
- Zoological Parks
 - Shankar Bagh Zoo- Junagadh- Asiatic Lion
 - P Naidu Zoo, Darjeeling : Red Panda, snow leopard
 - Rann of Katchch : Indian Wild Ass
- DNA/ Gene bank
 - CCMB- LaCONES
 - Center for conservation and molecular biology : Lab for conservation of endemic species.

Biodiversity Conservation



- Gene banks : storage of genetic components.
- They may use somatic cell nuclear transfer technology in case of low population of a species.
- Restore and Revival Programme :
 - Uses crispr Cas - 9 and biotech
 - Bring back the extinct woolly mammoth by modifying the genetic structure of the asian elephant

Wildlife Sanctuary	National Parks	Biosphere Reserves
Notified under WPA, 1972	Notified under WPA, 1972	As per UNESCO MAB programme
By state and centre in case of UT	By state and centre in few cases	Centre
Such area is of adequate ecological, faunal, floral, geomorphological, natural or zoological significance : particular species	Ecological, faunal, floral, geomorphological, or zoological association of importance. Not necessarily particular species.	To promote sustainable development based on local community efforts and sound science
567	106	18, 12 in World Network list (latest addition- Panna BR in 2020)
Limited biotic interference allowed	Strict prohibition	Divided into 3 zones- core, buffer, transition

In Situ Conservation



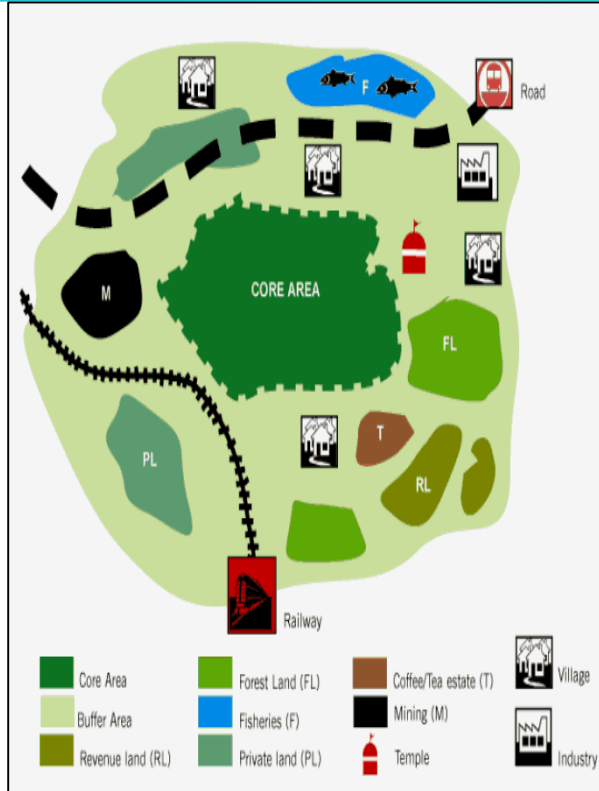
- Wildlife : concurrent list.
- National Parks : Human activity is greatly restricted.
- Tourism : Allowed in Wildlife Sanctuary, Allowed in Tiger Reserve but not allowed in core areas of biosphere reserves.

Tiger Reserves



- Tiger Reserves in India consist of a core tiger habitat, which would be managed as an inviolate area.
- These habitats are designated under the Wild Life Protection Act 1972.
- The Tiger Reserves in India must be preserved for the conservation of Tigers without compromising the rights of local tribes and other forest inhabitants.
- The National Tiger Conservation Authority is in charge of 54 Tiger Reserves in India.
- As per the reports, around 80% of tigers in the world are found in India and it becomes important for the government to protect these species.

Tiger Reserves



Tiger Reserves



- Tiger reserves in India are declared by the NTCA, constituted with two components.
 - Core Area– the core areas have the legal status of a national park or a sanctuary, exclusive tiger agenda is followed in the core areas.
 - Buffer Area- buffer or peripheral areas are a mix of forest and non-forest land, managed as a multiple use area, an inclusive people oriented agenda is followed in the buffer zone.

Tiger Reserves



- Project Tiger
- In order to protect our national animal, the tiger, India started the Project Tiger initiative in 1973.
- Project Tiger keeps an eye on 54 tiger reserves in India, which makes up roughly 2.21% of the nation's total land area.
- MoEFCC's Project Tiger is a centrally sponsored programme that offers tiger reserve states in India centralised support for tiger conservation.
- The NTCA oversees project tiger's execution.

NATIONAL TIGER CONSERVATION AUTHORITY (NTCA)



- It is a statutory body constituted under the WPA, 1972 for tiger conservation.
- It is headed by the minister of MOEFCC.
- The Authority has three Members of Parliament of whom two will be elected by the House of the People and one by the Council of States.
- It also has eight experts or professionals having qualifications and experience in wildlife conservation and welfare of people including Tribals.
- The Inspector General of Forests, in charge of project Tiger, is the ex-officio Member Secretary.

Elephant Reserves



- In order to protect elephants, their habitat, and migration routes, address issues relating to man-animal conflict and ensure the welfare of elephants kept in captivity, the Government of India established Project Elephant as a Centrally Sponsored Scheme in 1992.
- Elephant reserves were established towards this end.
- In 16 of India's 28 states, the Indian Elephant is frequently sighted, with a concentration in the Southern Western Ghats, North-Eastern India, Eastern India, Central India, and Northern India.
- There are 33 elephant reserves in India.

Elephant Reserves



- Elephant reserves are not notified under a particular law.
- According to the Wildlife (Protection) Act (WLPA) 1972, a 'protected area' can be one of a 'national park', a 'wildlife sanctuary', a 'conservation reserve' or a 'community reserve'. In the eyes of the law, an elephant reserve is no different from forest land or revenue land. The activities that are prohibited in protected areas – including mining, oil- and gas-drilling, dams, etc. – are permissible in an elephant reserve.
- In 2010, Elephant was designated as National Heritage Animal but nothing was done to preserve its sanctity in a legal sense.

Biosphere Reserves



- Promote solutions reconciling the conservation of biodiversity with its sustainable use.
- They are learning areas for sustainable development under diverse ecological, social and economic contexts, touching the lives of more than 250 million people.
- There are currently 738 biosphere reserves in 134 countries, including 22 transboundary sites, that belong to the World Network of Biosphere Reserves.

Biosphere Reserves



- The first biosphere reserve of the world was established in 1979
- India- 18 BR
 - 12 in World Network
 - Panna added in 2020
 - Encompasses:
 - Panna National Park,
 - Three sections of the Gangau wildlife Sanctuary (I, III and IV).
 - The Reserved Forests of the North Panna Forest Division.
 - Ken river flows through this terrain.
 - Ken-Betwa Link

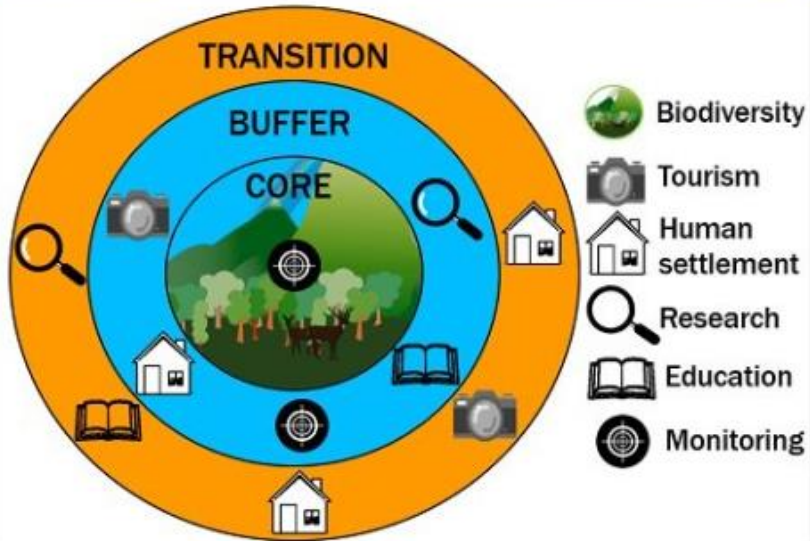
Biosphere Reserves



- Biosphere reserves : can contain national parks and wildlife sanctuaries within itself.
- Protection of both terrestrial and marine ecosystems and genetic diversity.
- A biosphere reserve has 3 zones.
- World Network of Biosphere reserves : interactive network of sites of excellence. Collaboration for sustainable development, knowledge sharing, poverty reduction, human well being, respect for cultural knowledge and enhanced capacity to cope with climate change.

Biosphere Reserves

Zones of Biosphere Reserves



Biosphere Reserves



The World Network of Biosphere Reserves is supported by different regional, sub-regional or thematic networks:

The **African Biosphere Reserves Network (AfriMAB)** was created in 1996 and consists of 33 African countries. It is one of the means to implement UNESCO's Global Priority on Africa.

The **ArabMAB Network** was officially launched in 1997 and represents 18 Arab countries.

The **East Asian Biosphere Reserve Network** was launched in 1994. Today, it consists of China, the Democratic People's Republic of Korea, Japan, Kazakhstan, Mongolia, the Republic of Korea and the Russian Federation.

EuroMAB comprises all the biosphere reserves in Europe and North America. Created in 1987, it is currently the largest MAB Regional Network with 53 countries.

The **Ibero-American MAB Network (IberoMAB)** was created in 1992. It comprises 22 countries from Latin American and the Caribbean, Portugal and Spain.

The **Pacific Man and the Biosphere Network (PacMAB)** was created in 2006 and consists of the Federated States of Micronesia, Kiribati, Palau, Papua New Guinea, Samoa and Tonga.

The **South and Central Asia MAB Network (SACAM)** was created in 2002 and comprises Afghanistan, Bangladesh, Bhutan, India, Iran, Kazakhstan, Kyrgyzstan, Maldives, Nepal, Pakistan, Tajikistan, Sri Lanka and Uzbekistan.

The **Southeast Asian Biosphere Reserve Network (SeabRnet)** was created in 1998. Today, it includes Cambodia, China, Indonesia, Japan, Lao PDR, Malaysia, Myanmar, Philippines, Thailand, Timor-Leste and Viet Nam.

The **World Network of Island and Coastal Biosphere Reserves** was established in 2012 and includes 36 countries. It aims to study, implement and disseminate island, marine and coastal strategies to preserve biodiversity and heritage, promote sustainable development, and adapt to and mitigate the effects of climate change.

The **Mediterranean Biosphere Reserves Network** guides the strengthening of relations between biosphere reserves in the Mediterranean. It aims to build bridges for dialogue and collaboration and foster research into major environmental issues affecting the shores of the Mediterranean. Currently, around 70 biosphere reserves in 15 countries belong to this Network.



Meeting global challenges and creating sustainable and long-term impacts is only possible through collaboration and broad partnerships. MAB works together with other UN agencies, international and national partners, governments, NGOs, academia and the private sector, and works to promote North-South, South-South and South-North-South cooperation. Examples of such partnerships include:

The **Great Apes Survival Partnership (GRASP)** brings together 95 partners including UN agencies, national governments, intergovernmental organizations, conservation organizations and private companies. It aims to remove the threat of imminent extinction faced by gorillas, chimpanzees, bonobos and orangutans.

The **Spanish Cooperation (AECID)** and the **Spanish Autonomous Authority for National Parks (OAPN)** are funding a reforestation project in La Selle Biosphere Reserve in Haiti to contribute to the sustainable human and economic development of this site.

The **Islands of Jeju (Republic of Korea)** and **Menorca (Spain)** support and finance the activities of the World Network of Island and Coastal Biosphere Reserves.

In the framework of a 5-year agreement, the **Belgium Science Policy Office (BELSPO)** is supporting the MAB programme and biosphere reserves in Africa, particularly with scientific studies.

The **Government of Flanders (Belgium)** is financing a project implemented by young people from Latin American and Caribbean biosphere reserves who are demonstrating through concrete activities how these sites can be a tool to improve the socio-economic development of youth.

The **African Development Bank** is funding the **BIOALP** project, which aims to strengthen the capacities of the Member States of the Lake Chad Basin Commission – Cameroon, Chad, Central African Republic, Niger and Nigeria – to safeguard and sustainably manage the hydrological, biological, and cultural resources of the Lake Chad basin across their borders, in order to support poverty reduction and promote peace.

In 2019, the programme signed an agreement with **LVMH** to work on sustainability projects in Biosphere Reserves around the world.



The MAB programme contributes to global efforts for education and capacity-building through workshops, training courses, educational programmes and partnerships with professional and educational institutions, including the following:

The **UNESCO Category II Centre Regional School on Integrated Management of Tropical Forests and Territories (ERAFT)** at the University of Kinshasa (Democratic Republic of Congo) is the first MAB Category II Centre in Africa. It trains about 60 experts from French and Portuguese-speaking countries in Africa each year.

The **UNESCO Category II International Centre on Mediterranean Biosphere Reserves** in Castellet I la Corna, Spain is funded by the Abertis Foundation. It provides an excellent platform for cooperation on issues related to biosphere reserves for all Mediterranean countries.

There are now 48 **University Twinning and Networking Programme/UNESCO Chairs** related to biosphere reserves and sustainable development.

Training Courses for Island and Coastal Biosphere Reserves Managers. Since 2013, Jeju Island (Republic of Korea) and the MAB Programme have provided annual training courses for managers of island and coastal biosphere reserves.

The MAB Programme also recognizes outstanding work in biosphere reserves through a variety of awards:

The **MAB Young Scientists Award** is granted annually to young researchers to support research into ecosystems, natural resources and biodiversity.

The **UNESCO Sultan Qaboos Prize for Environmental Preservation** recognizes outstanding contributions in the management or preservation of the environment, consistent with the policies, aims and objectives of UNESCO and the Organization's programmes in this field.

The **Michel Batisse Award** is presented every two MAB International Coordinating Councils in memory of Dr. Michel Batisse. It recognizes excellence in the management of biosphere reserves.



The MAB Programme is an intergovernmental scientific programme that aims to establish a scientific basis for enhancing the relationship between people and their environments. It combines the natural and social sciences with a view to improving human livelihoods and safeguarding natural and managed ecosystems, thus promoting innovative approaches to economic development that are socially and culturally appropriate and environmentally sustainable.

OUR VISION

A world where people are conscious of their common future and their interactions with the planet, and act collectively and responsibly to build thriving societies in harmony within the biosphere. The MAB programme and its World Network of Biosphere Reserves (WNBR) serve this vision through biosphere reserves and beyond.

OUR MISSION FOR 2015-2025

- Develop and strengthen models of sustainable development through the WNBR;
- Communicate experiences and lessons learned, and facilitate the global diffusion and application of these models;
- Support evaluation and high-quality management of biosphere reserves, strategies and policies for sustainable development and planning, and accountable and resilient institutions;
- Help Member States and stakeholders to achieve the Sustainable Development Goals by sharing experiences and lessons learned related to exploring and testing policies, technologies and innovations for the sustainable management of biodiversity and natural resources and mitigation and adaptation to climate change.




MAN AND THE BIOSPHERE PROGRAMME
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Biosphere Reserves



HOW DOES MAB WORK?

UNESCO's intergovernmental structure provides the MAB programme with a framework to support national governments in the planning and implementation of research and training programmes, through technical assistance and scientific advice.

Participating countries are invited to establish MAB National Committees, that ensure national participation in the international programme, by defining and implementing country's priorities and activities. The MAB programme currently operates through 158 MAB Programme National Committees established among the 195 Members States and nine Associate Members States of UNESCO.

The agenda of the MAB Programme is defined by its main governing body, the International Coordinating Council. The MAB Council consists of 34 Member States elected by UNESCO's General Conference. The Council elects a chair and five vice-chairpersons from each of UNESCO's geopolitical regions, one of which functions as a rapporteur. These constitute the MAB Bureau.

The MAB Secretariat is based at UNESCO's Division of Ecological and Earth Sciences, and works closely with the different UNESCO Field Offices around the world to coordinate the work of the MAB Programme at national and regional levels. Its staff members draw on expertise in many and varied disciplines.

MAB is funded through the regular budget of UNESCO and mobilizes funds-in-trust granted by Member States, bilateral and multilateral sources, and extra-budgetary funds provided by countries, the private sector and private institutions. The Programme can grant seed funding to assist countries in developing projects and/or to secure appropriate partnership contributions.

A NEW ROADMAP FOR MAB

To ensure a strong response to contemporary development challenges and opportunities in the coming years, the MAB Council adopted the MAB Strategy 2015-2025 and the Lima Action Plan 2016-2025. Together, they constitute a roadmap to help implement the UN 2030 Agenda and its Sustainable Development Goals (SDGs), and the Paris Agreement on climate change.

The MAB Strategy focuses on supporting Member States and stakeholders to conserve biodiversity, restore and enhance ecosystem services, and foster the sustainable use of natural resources. The Lima Action Plan includes targeted outcomes, actions and outputs that will contribute to the effective implementation of the Strategic Objectives contained in the MAB Strategy. It also specifies the main entities responsible for implementation, together with time frames and performance indicators.

Process of Excellence and Enhancement

In May 2013, the MAB Council (MAB-ICC) adopted the Exit Strategy. Its purpose is to improve the credibility and quality of the World Network of Biosphere Reserves and to help Member States establish the required standards for their biosphere reserves, in order to become fully functional and conform to criteria set by the Statutory Framework for Biosphere Reserves. All concerned Member States engaged in this process and demonstrated their commitment to the MAB programme and the WBNBR.

In June 2017, MAB-ICC agreed to finalize the Exit Strategy by 2020 and to institute a 'Process of excellence and enhancement of the WBNBR as well as quality improvement of all members of the World Network'. This aims to ensure that the WBNBR will serve as models for implementation of the SDGs.

WHAT ARE BIOSPHERE RESERVES?

Biosphere reserves are 'learning places for sustainable development'. They are sites for testing interdisciplinary approaches to understanding and managing changes and interactions between social and ecological systems, including conflict prevention and management of biodiversity and climate change. They are places that provides local solutions to global challenges.

Biosphere reserves include terrestrial, marine and coastal ecosystems. Each site promotes solutions reconciling the conservation of biodiversity with its sustainable use.

Biosphere reserves are nominated by national governments and remain under the sovereign jurisdiction of the states where they are located. Their status is internationally recognized.

They consist of three interrelated zones that aim to fulfil three complementary and mutually reinforcing functions:

- ✓ The core area(s) comprises a strictly protected zone that contributes to the conservation of landscapes, ecosystems, species and genetic variation.
- ✓ The buffer zone surrounds or adjoins the core area(s), and is used for activities compatible with sound ecological practices that can reinforce scientific research, monitoring, training and education.
- ✓ The transition area is where communities foster socio-culturally and ecologically sustainable economic and human activities.

WORLD NETWORK OF BIOSPHERE RESERVES (WBNBR)

The WBNBR consists of a dynamic and interactive network of sites of excellence. It works to foster the harmonious integration of people and nature for sustainable development through participatory dialogue, knowledge sharing, poverty reduction, increased human well-being, respect for cultural values and enhanced capacities to cope with climate change. It promotes North-South, South-South and South-North-South collaboration and represents a unique tool for international cooperation through the exchange of experiences and know-how, capacity-building and the promotion of best practices.

The WBNBR is a network of socio-ecosystems dedicated to interdisciplinary research, capacity-building, and management and experimentation that comprises innovative combinations of economic, environmental, social and energy alternatives for sustainable development.

There are 701 biosphere reserves in 124 countries, including 21 transboundary sites*.

79 in 29 countries in Africa

33 in 12 countries in the Arab States

157 in 24 countries in Asia and the Pacific

302 in 38 countries in Europe and North America

130 in 21 countries in Latin America and the Caribbean

*as of August 2019



Young people are playing a key role in helping the MAB Programme and its World Network in its effort to build thriving societies in harmony with the biosphere. It all started with the First MAB Youth Forum held in Italy, in September 2017, with hundreds of delegates from 85 different countries. The follow-up regional event, the First IberoMAB Youth Forum, took place in Ecuador in December 2019, with participants from Latin America, Caribbean, Portugal and Spain. Several countries are organizing regional or national forums. They have also created their youth networks to ensure the implementation of several initiatives.

The 2nd International MAB Youth Forum was held in the Changbaishan Biosphere Reserve (China) in September 2019. A large MAB Youth delegation attended the Youth Climate Summit, at UN New York on 21 September 2019 to showcase their climate actions and exchange ideas.



THE TOTAL TERRESTRIAL AND MARINE AREA COVERED BY BIOSPHERE RESERVES AROUND THE WORLD AMOUNTS 6,812,000 KM²; ALMOST THE SIZE OF AUSTRALIA. AROUND 257 MILLION PEOPLE LIVE IN BIOSPHERE RESERVES.

Community and Conservation Reserve



Community Reserve

Areas outside NPs/WLSs, act as buffer, inhabited by people

Siswan (Punjab), Max in Nagaland

Conservation Reserve

Are uninhabited and completely owned by the Centre;

Beas River (Punjab), Asan (Uttarakhand)

Both declared under WPA, 1972 (2002 amendment)

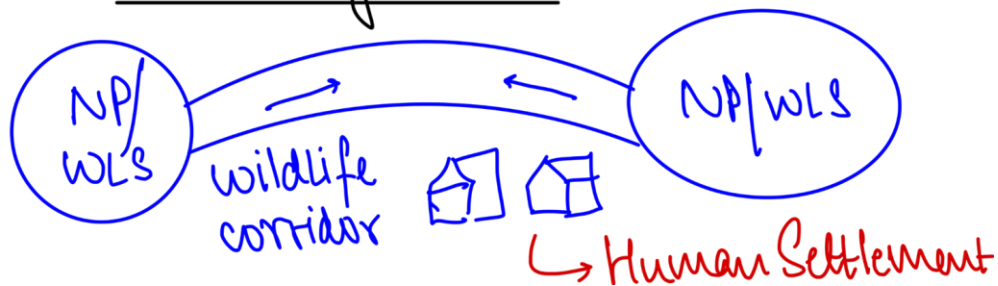
Community and Conservation Reserve



- Community Reserve : Area outside NP /WLS that act as a connector to other NP /WLS. Privately owned land can be declared community reserve. Here, protection is done with the help of the community.
- Conservation reserve : area contiguous to existing protected area / wildlife corridor where local communities use land for livelihood. This area is owned by the government.

Community and Conservation Reserve

Community Reserve



Conservation Reserve

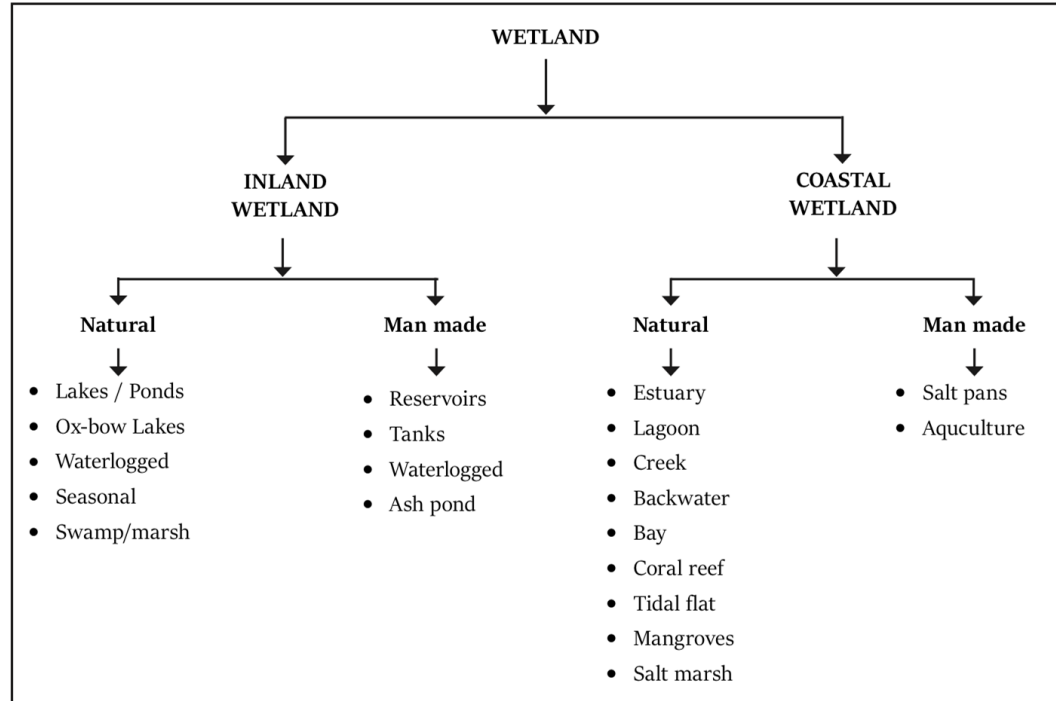


- Definition
 - Areas of marsh, fen, peatland/water, whether natural (or) artificial, permanent (or) temporary with water that is static (or) flowing, fresh, brackish (or) salt, including areas of marine water the depth of which at low tide does not exceed 6 mtrs.
- Characteristics
 - Covered by water (or) has waterlogged soil for atleast seven days during the growing season.
 - Adopted plant life (hydrophytes)
 - Hydric soils (not enough O₂ available for some plants)

- Functions of Wetlands
 - Habitat to aquatic flora and fauna, as well as numerous species of birds, including migratory species.
 - Filtration of sediments and nutrients from surface water
 - Nutrients recycling
 - Water purification
 - Floods mitigation
 - Maintenance of stream flow
 - Ground water recharging
 - Provide drinking water, fish, fodder, fuel, etc

- Reasons for depletion
 - Conversion of lands for agriculture
 - Overgrazing
 - Removal of sand from beds
 - Aqua culture
 - Habitat Destruction and Deforestation
 - Pollution
 - Domestic waste

Wetlands

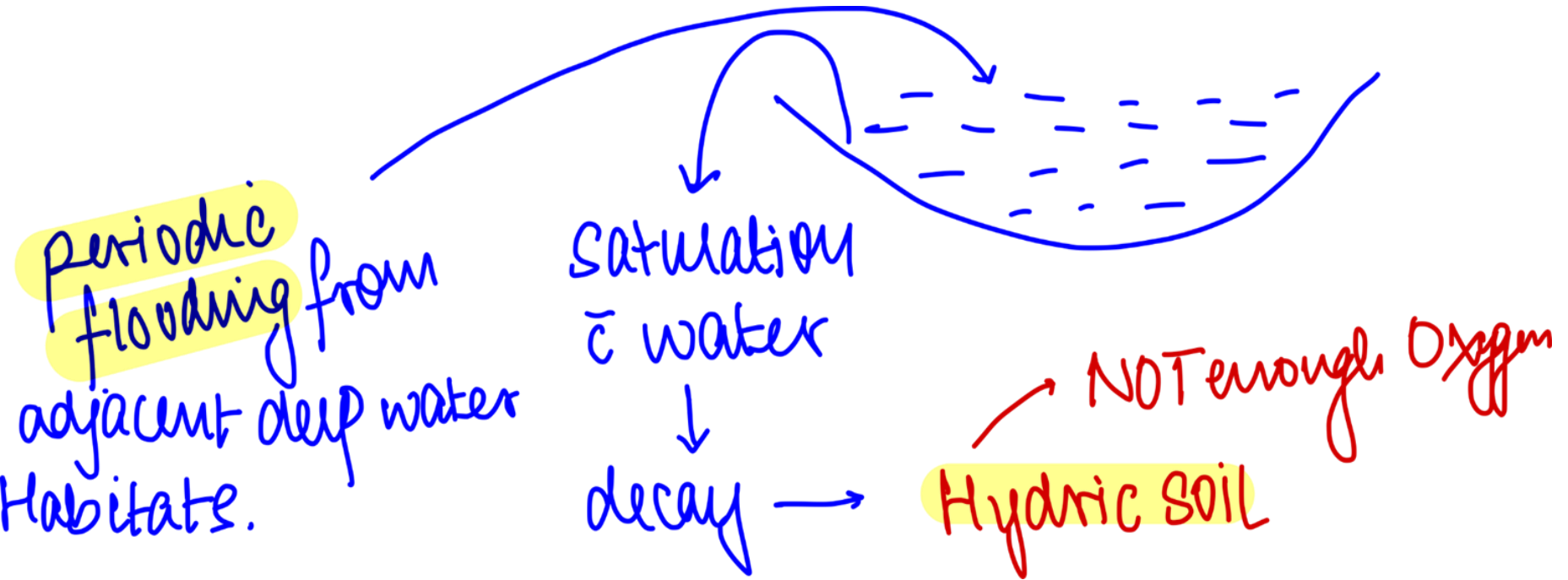


Wetlands



- Wetlands have upper soils saturated with water, so it leads to anaerobic conditions leading to decay of organic matter.
- Periodic flooding from neighboring areas leads to saturation with water.
- There is not enough oxygen in the soil : known as hydric soils.

Wetlands



Wetlands and lakes

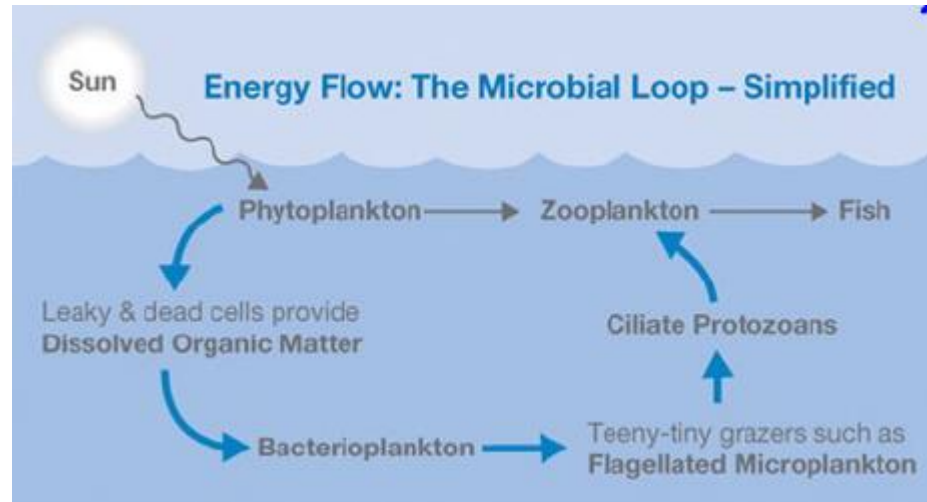


- National Lake Conservation Programme (NLCP) considers lakes as standing water bodies which have a minimum water depth of 3 m, generally cover a water spread of more than ten hectares, and have no or very little aquatic vegetation (macrophytes).
 - Use of lakes- drinking water supplies, irrigation and/or recreation
 - Hence, excessive growth of macrophytes affects the water quality adversely and interfere with the utilization of the water body.
- Wetlands (generally less than 3 m deep over most of their area), on the other hand, are usually rich in nutrients (derived from surroundings and their sediments) and have abundant growth of aquatic macrophytes.
 - They support high densities and diversity of fauna, particularly birds, fish and macro invertebrates, and therefore, have high value for biodiversity conservation
- Lakes are generally less important when compared to wetland from the viewpoint of ecosystem and biodiversity conservation.

Wetlands vs Lakes

Characteristics	Lake	Wetland
Origin	Various process	Fluvial/ Geomorphic processes
Water Turnover	Permanent	Permanent or Temporary
Water level changes	Relatively small	Relatively large
Littoral: Pelagial ratio	Small	Large
Thermal stratification	Yes	No
Vertical mixing	Thermally regulated	Wind regulated
Dominant Producer	Phytoplankton	Macrophytes
Food chain	Grazing pathway	Detritus pathway
Nutrient Cycles	Microbial loop less important	Microbial loop dominant
Productivity	Low	High
Trophic status	Oligotrophic	Mostly Eutrophic
Biodiversity	Generally low	Generally high
Management Objectives	Control of eutrophication, high water quality	Biodiversity conservation

Wetlands and lakes



Wetland (Conservation and Management) Rules 2017



- MoEFCC notified these rules in 2017 under Environment protection act, 1986
- Key points of new Rules:
- State Wetlands Authority (SWA):
- It stipulates setting up of SWA in each State/UTs headed by State's environment minister and include range of government officials and experts.

Wetland (Conservation and Management) Rules 2017



- Digital inventory of all wetlands:
 - Mandatory for state authorities to prepare list of all wetlands and list of wetlands to be notified within six months.
 - Based on it, a comprehensive digital inventory of all wetlands will be created and will be updated every ten years.

Wetland (Conservation and Management) Rules 2017



- National Wetlands Committee (NWC):
 - The rules stipulates for setting up of NWC, headed by MoEFCC Secretary, to monitor implementation of these rules and oversee work carried out by States.
 - NWC will also advise Central Government on appropriate policies and action programmes for conservation and wise use of wetlands, recommend designation of wetlands of international importance under Ramsar Convention, advise on collaboration with international agencies on issues related to wetlands etc.

- Ramsar Convention is a convention on wetlands that was signed in 1971 in the Iranian city of Ramsar
 - The official name of the treaty is The Convention on Wetlands of International Importance, especially as Waterfowl Habitat.
- The convention works on three pillars that define the purpose of the Ramsar Convention:
 - Wise Use – To work towards the wise use of all wetlands
 - List of Wetlands of International Importance – Designate suitable wetlands under the Ramsar List to effectively manage those
 - International Cooperation – To bring cooperation internationally over the transboundary wetlands, shared wetland systems and shared species.

- Montreux Record
 - It is a mechanism that was launched in 1990 and is associated with the Ramsar Advisory Mission. It is a register of the list of those Ramsar Sites that need urgent attention.
 - Chilika Lake in Orissa was the first to be inscribed on the Montreux Record in 1993 owing to siltation restricting the lake's mouth.
 - It was taken off the Record in 2002 after the government's restoration efforts.

RAMSAR Sites : current status



Currently, two wetlands of India are in Montreux record: Keoladeo National Park (Rajasthan) and Loktak Lake (Manipur).

- Currently 75 wetlands in India are designated as Ramsar sites

The Ramsar Sites Criteria

The nine criteria for identifying Wetlands of International Importance

Group A of the Criteria. Sites containing representative, rare or unique wetland types

Criterion 1: A wetland should be considered internationally important if it contains a representative, rare, or unique example of a natural or near-natural wetland type found within the appropriate biogeographic region.

Group B of the Criteria. Sites of international importance for conserving biological diversity

Criteria based on species and ecological communities

Criterion 2: A wetland should be considered internationally important if it supports vulnerable, endangered, or critically endangered species or threatened ecological communities.

Criterion 3: A wetland should be considered internationally important if it supports populations of plant and/or animal species important for maintaining the biological diversity of a particular biogeographic region.

Criterion 4: A wetland should be considered internationally important if it supports plant and/or animal species at a critical stage in their life cycles, or provides refuge during adverse conditions.

Specific criteria based on waterbirds

Criterion 5: A wetland should be considered internationally important if it regularly supports 20,000 or more waterbirds.

Criterion 6: A wetland should be considered internationally important if it regularly supports 1% of the individuals in a population of one species or subspecies of waterbird.

Specific criteria based on fish

Criterion 7: A wetland should be considered internationally important if it supports a significant proportion of indigenous fish subspecies, species or families, life-history stages, species interactions and/or populations that are representative of wetland benefits and/or values and thereby contributes to global biological diversity.

Criterion 8: A wetland should be considered internationally important if it is an important source of food for fishes, spawning ground, nursery and/or migration path on which fish stocks, either within the wetland or elsewhere, depend.

Specific criteria based on other taxa

Criterion 9: A wetland should be considered internationally important if it regularly supports 1% of the individuals in a population of one species or subspecies of wetland-dependent nonavian animal species.

Environmental Legislations



- Wildlife Protection Act, 1972
- Indian Forest Act, 1927
- The Forest (Conservation) Act of 1980
- Forest Act, 2006
- Environment Protection Act, 1986
- Bio Diversity Act, 2002
- The Water (Prevention and Control of Pollution) Act of 1974

Environmental Legislations



- Water (Prevention and control of Pollution) Cess Act, 1977
- The Air (Prevention and Control of Pollution) Act of 1981
- National Green Tribunal Act 2010
- Compensatory Afforestation Act, 2016

Constitutional Provisions



- 42nd CAA, 1976
- Article 51A (g) – FUNDAMENTAL DUTY
 - “It shall to be duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wild life and have compassion for living creatures.”
- Article 48A - DIRECTIVE PRINCIPLES OF STATE POLICY
 - “The State shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country”.

Constitutional Provisions



- Deletion of Entry 19 from List II of the Schedule(State list)
 - A new entry (Entry 17-A) related to forests was inserted in the Concurrent list

Wildlife Protection Act, 1972



- Provides for:
- Appointment of Wildlife Advisory Board, Wildlife Warden, their powers, duties
- Comprehensive listing of endangered wildlife species was done for the first time
- Prohibition of hunting of the endangered species was mentioned.
- Regulations for hunting wild animals and birds
- Establishment of Sanctuaries, National Parks, community and conservation reserves
- Regulations for trade in wild animals, Animal products and trophies
- Judicially imposed penalties for violating the act

Wildlife Protection Act, 1972



- **Authorities established** under this act
- **Wildlife Crime Control Bureau (statutory status)**– headed by Director of Wildlife Preservation – appointed by Central Government
- **Chief Wildlife Warden** – state Governments
- **Central Zoo Authority**
- **National Board of Wildlife** – headed by Prime Minister
- **State Board of Wildlife** – headed by Chief Minister
- **National Tiger Conservation Authority** – headed by minister of MOEFCC
- Tiger conservation foundation – To be established by states

Wildlife Protection Act, 2021 Amendments



- Implement provisions of Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)
 - Central government to designate a
 - (i) **Management Authority**, which grants export or import permits for trade of specimens
 - (ii) **Scientific Authority**, which gives advice on aspects related to impact on the survival of the specimens being traded.
 - Every person engaging in trade of a scheduled specimen must report the details of the transaction to the Management Authority.
 - Identification mark to be used by the MA for a specimen, as per CITES.

Wildlife Protection Act, 2021 Amendments



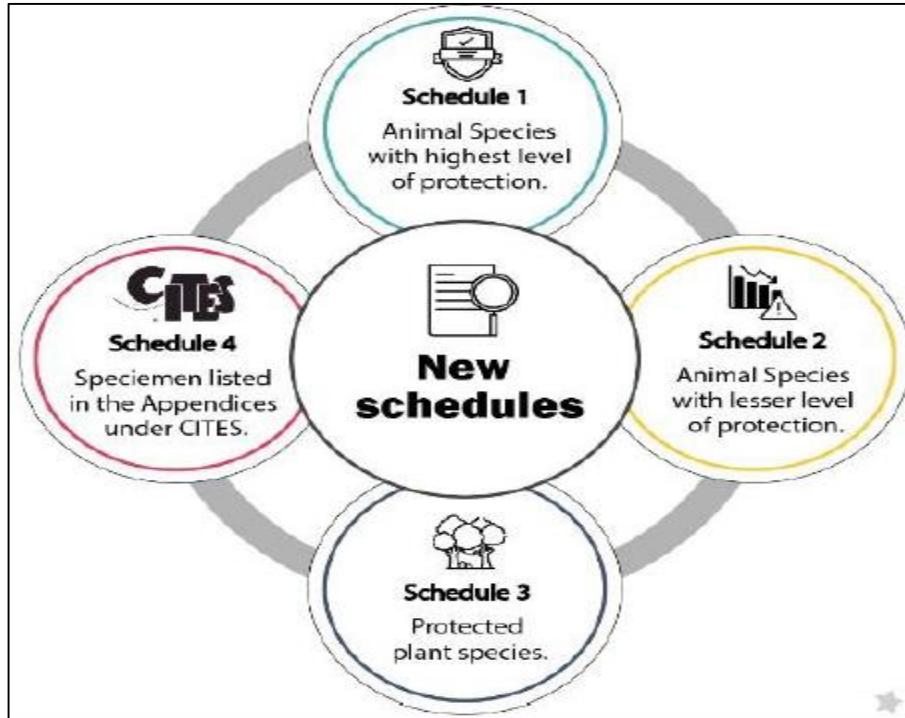
- Earlier, WPA had 6 schedules
- Schedule 1 and 2 : animals given absolute protection from hunting as they were in the category of endangered species.
- Schedule 3 and 4 : Similar provisions, but animals were not in the danger of becoming extinct.
- Schedule 5 : vermins like ducks and deer that could be hunted with prior permission of chief wildlife warden
- Schedule 6 : cultivation of plant life.

Wildlife Protection Act, 2021 Amendments



- Rationalization of schedules
 - Earlier- 6 schedules
 - Now- 4
 - **Schedule 1**- Animal Species with highest level of protection
 - **Schedule 2**- Animal Species with lower level of protection
 - **Schedule 3**- Protected Plant species
 - **Schedule 4**- Specimen listed under CITES Appendices

Wildlife Protection Act, 2021 Amendments



Wildlife Protection Act, 2021 Amendments



- Wild animals will be declared as Vermin by Central Government by the way of notification for any area and for a specified period.
- Chief Wildlife Warden will manage and protect sanctuaries in accordance with the management plans prepared as per guidelines of the **central** government.
- In cases where FRA, 2006 is applicable, the management plan will be prepared after consultation with gram sabha of that area.
- Increase in penalties

Wildlife Protection Act, 2021 Amendments



- Exemptions for live elephants
- Transfer or transport of live elephants allowed for a religious or any other purpose by person having ownership certificates in accordance with conditions prescribed by the Central Government.
- Central government empowered to regulate or prohibit the import, trade, possession or proliferation of invasive alien species.
- State Board for Wild Life permitted to constitute a Standing Committee

Wildlife Protection Act, 2021 Amendments



- Surrender of captive animals : any person having a certificate of ownership for captive animals can voluntarily surrender them to chief wildlife warden.
- No compensation to be paid for person surrendering such items.
- Permits may be granted for film making.
- Certain activities by local communities allowed without a permit in a sanctuary.
- No renewal of arms licenses to be granted to any person residing within 10 km of a sanctuary except by permission of CWLW.

Forest Protection Laws



- **Indian Forest Act, 1927**

- Regulation of forests by government
- Classified into timber and non-timber
- Permission needed to harvest timber from forest land
- Regulation of rights of local communities to use Minor Forest Produce

Forest Protection Laws



- Reserved forests : no human activity allowed. 54.4% of all forests.
- Protected forests- Forests where local communities are allowed certain rights like collecting firewood and grazing animals. 45.6% of total forests
- Both reserved and protected forest are notified by the state government.
- Degree of protection
 - Reserved forests > Protected forests > Village forests
- Recent amendments- bamboo= MFP but can be used only outside forest area

Forest Protection Laws



- **Forest Conservation Act, 1980**
- To provide for the conservation of forests and for matters connected to it. To protect, conserve and manage forests.
- Prior approval of the Central Government is essential for diversion of forest lands for the non-forestry purposes.
- Compensation needs to be there for loss of forests due to non forestry economic activity to be paid by forest developer.

Forest Protection Laws



- **Forest Conservation Act, 1980**
- Deals with deforestation and afforestation and loss of forests due to economic activity.
- Compensation mechanism is as per the compensatory afforestation act.

Forest Protection Laws



- **Compensatory Afforestation Fund Act, 2016**
- The Compensatory Afforestation Fund Management and Planning Authority (**CAMPA**) Act or **Compensatory Afforestation Fund Act** seeks to provide an appropriate institutional mechanism, **both at the Centre and in each State and Union Territory**, to **ensure expeditious utilization in the efficient and transparent manner of amounts released in lieu of forest land diverted for the non-forest purpose** which would mitigate the impact of diversion of such forest land.

Forest Protection Laws



- The **CAF Act** was passed by the center in **2016** and the related rules were notified in **2018**.
- It seeks to establish the **National Compensatory Afforestation Fund** under the **Public Account of India**, and a State Compensatory Afforestation Fund under the Public Account of each state.
- As per the rules, **90% of the CAF money is to be given to the states while 10% is to be retained by the Centre**.

Forest Protection Laws



- These Funds will receive payments for
 - compensatory afforestation,
 - Net present value of forest (NPV), and
 - other project specific payments.

HOW CAMPA WORKS?



for every tree : Net present value (N.P.V.)

eg: Rs 100

total → 10,000

↓
Central CAMPA
v/ MOEFCC.

₹9,000

to State
CAMPA

→ will ascertain
where to plant
trees.

↓
₹200

on
afforestation

₹1800

on auxiliary
activity

↓
equivalent area must
be brought v/ forest cover

Forest Protection Laws



The funds can be **used for**:

1. treatment of catchment areas,
2. Assisted natural generation,
3. Forest management,
4. Wildlife protection and management,
5. Relocation of villages from protected areas,
6. Managing human-wildlife conflicts,
7. Training and awareness generation,
8. Supply of wood saving devices, and
9. Allied activities.

- **Forest Rights Act, 2006**

- Implemented by Ministry of Tribal Affairs
- The act recognizes and vest the forest rights and occupation in Forest land in forest Dwelling Scheduled Tribes (FDST) and Other Traditional Forest Dwellers (OTFD) who have been residing in such forests for generations.

Forest Protection Laws



- The act identifies four types of rights:
- **Title rights**
 - It gives FDST and OTFD the right to ownership to land farmed by tribals or forest dwellers subject to a maximum of 4 hectares.
 - Ownership is only for land that is actually being cultivated by the concerned family and no new lands will be granted.
 - Cut off date : 13 december, 2005

Forest Protection Laws



- **Use rights**
 - The rights of the dwellers extend to extracting Minor Forest Produce, grazing areas, to pastoralist routes, etc.
- **Relief and development rights**
 - To rehabilitation in case of illegal eviction or forced displacement and to basic amenities, subject to restrictions for forest protection

Forest Protection Laws



- **Forest management rights**

- It includes the right to protect, regenerate or conserve or manage any community forest resource which they have been traditionally protecting and conserving for sustainable use.

Forest Protection Laws



- India State of Forest Report (ISFR), 2021
- Assessment of India's forest and tree cover, published every two years by the Forest Survey of India under MoEFCC
- ISFR 2021 is the 17th report.
 - The Total Forest and Tree cover is 24.62% of the geographical area of the country.
 - The Total Forest cover is 7,13,789 sq km which is 21.71% of the geographical area of the country.
 - The Tree cover is 2.91% of the geographical area of the country.

Forest Protection Laws



- Definition of forest (ISFR)
- The forest cover is defined as 'all land, more than one hectare in area, with a tree canopy density of more than 10 percent irrespective of ownership and legal status. Such land may not necessarily be a recorded forest area. It also includes orchards, bamboo and palm'
- Forest cover figures are divided as 'Inside Recorded Forest Area' and 'Outside Recorded Forest Area'.
- Those 'Inside Recorded Forest Area' are basically natural forests and plantations of Forest Department.
- The Forest cover 'Outside Recorded Forest Area' are mango orchards, coconut plantations, block plantations of agroforestry.

Forest Protection Laws



Forest Protection Laws



- Largest forest cover in India: Madhya Pradesh > Arunachal Pradesh > Chhattisgarh > Odisha > Maharashtra
- Forest density: Mizoram (84.53%) > Arunachal Pradesh (79.33%) > Meghalaya (76.00%) > Manipur (74.34%) > Nagaland (73.90%).
- States showing a significant gain in forest cover: Andhra Pradesh > Telangana > Odisha > Karnataka > Jharkhand.
- States showing loss in forest cover: Arunachal Pradesh > Manipur > Nagaland > Mizoram > Meghalaya.

- **New York Declaration**

- The New York Declaration on Forests is a voluntary and non-legally binding political declaration which grew out of dialogue among governments, companies and civil society, spurred by the United Nations Secretary-General's Climate Summit held in New York in 2014.
- The Declaration pledges to halve the rate of deforestation by 2020, to end it by 2030, and to restore hundreds of millions of acres of degraded land.
- Presently, India has not signed the New York Declaration on Forests (NYDF).

Forest Related Information



- **‘Tree City of the World’ (TCW) Tag**

- Mumbai and Hyderabad have been jointly recognised as ‘2021 TCW’.
- TCW programme has been started by United Nations’ Food and Agriculture Organisation and American non-profit organisation Arbor Day Foundation.
- Provides direction, assistance, and worldwide recognition for a community’s dedication to its urban forest and framework for healthy, sustainable urban forestry programme in town or city.
- City was evaluated based on five standards i.e. Establish Responsibility, Set the Rules, Know What You Have, Allocate the Resources, and Celebrate the Achievements.

- **Nagar Van Scheme**

- Launched in 2020, on World Environment Day (5th June) to develop 400 Urban Forests (Nagar Van) and 200 Nagar Vatika across the country in the next five years.
- Under the scheme
- A minimum of 20 hectares of forests will be created in the city.
- Forests will come up either on existing forest land or on any other vacant land offered by urban local bodies.
- Van Udyan once established will be maintained by the State Government.
- Fully funded by the under CAMPA (Compensatory Afforestation Fund Act, 2016) funds.

- **Miyawaki Method**

- It is a technique of urban afforestation by creating micro forests over small plots of land.
- Devised by Japanese botanist Akira Miyawaki in 1980s.
- Ensures 10 times faster growth of plant and 30 times denser than usual.
- In this technique, native trees of region are divided into four layers (shrub, sub-tree, tree, and canopy) after identification and analysis of soil quality.

Water and Air Acts



- THE WATER (PREVENTION AND CONTROL OF POLLUTION) ACT OF 1974
 - Prevention and control of water pollution and maintaining or restoring of wholesomeness and purity of water.
 - Act vests regulatory authority in State Pollution Control Boards and empowers these Boards to establish and enforce effluent standards for factories discharging pollutants into water bodies.
 - A Central Pollution Control Board performs the same functions for Union Territories and formulate policies and coordinates activities of different State Boards.

Water and Air Acts



- Water act 1974 was amended in 1988 to allow the SPCB and CPCB to close a defaulting industrial plant.

- THE AIR (PREVENTION AND CONTROL OF POLLUTION) ACT OF 1981
 - Enacted to implement the decisions taken in the stockholm convention.
 - The Air Act's framework is similar to that of the Water Act of 1974.
 - The Air Act expanded the authority of the central and state boards established under the Water Act, to include air pollution control.
 - The 1987 amendment introduced a citizen's suit provision into the Air Act and extended the Act to include noise pollution.

Environment Protection Act, 1986



- The Act empowers the Central Government to take necessary actions for preventing environmental pollution in all its forms and to tackle specific problems that are peculiar to different parts of the country.
- It was enacted in the wake of Bhopal Gas Tragedy
- Act specifically empowers State Government to designate air pollution areas and to prescribe the type of fuel to be used in these designated areas.

Environment Protection Act, 1986



- EPA was enacted under Article 253 of the constitution.
- The purpose of the Act is to act as an “umbrella” legislation designed to provide a framework for Central government co-ordination of the activities of various central and state authorities established under previous laws, such as Water Act and Air Act.

Environment Protection Act, 1986



- Sec (3) - Empowers government to constitute authority/s for the purpose of exercising powers and performing such functions (as conferred by the Act)
 - Central Groundwater Authority
 - Taj Trapezium Zone Pollution (Prevention and Control) Authority 2003
 - Environment Pollution(Prevention and Control) Authority for the NCR
- All the important legislations like Plastic, e-waste etc Waste Management Rules are issued under this Act
- ADD

Environment Protection Act, 1986



- Ecologically Sensitive Zones
- Ecologically important and fragile areas around protected areas
- Notified by: The Central Government in exercise of the powers conferred by section 3 of the EPA, 1986.
- Extent of ESZs as per 2011 guidelines: Generally, width of up to 10 kms around a PA.
- Create some kind of “Shock Absorber” for the Protected Areas (PAs).
- Recently SC directed every protected forest, national park and wildlife sanctuary in the country should mandatorily have a minimum 1 km ESZ,

Environment Protection Act, 1986

CATEGORIES OF ACTIVITIES UNDERTAKEN WITHIN AN ESZ



Prohibited

Commercial mining, setting of saw mills, polluting industries, major hydroelectric projects etc.



Regulated

Felling of trees, Establishment of hotels and resorts, drastic change of agriculture system, widening of roads, introduction of exotic species etc.



Permissible

Rainwater Harvesting, Organic farming, Ongoing Agricultural and Horticulture practices by local communities, Adoption of green technology for all activities etc.

PROCEDURE OF DECLARATION OF ESZ



Creation of inventory

- Inventory of activities around each of the Protected Area as well as important Corridors is made with the help of range officers.



Constitution of a committee

- The committee is tasked with demarcating extent of ESZ and create a Master Plan for its management (including classification of activities to be regulated).
- The committee generally comprises of: Wildlife Warden, Warden, an Ecologist and officials from local self-government and Revenue Department.

ESZ

Notification of ESZ

- Final proposal is forwarded to the MoEFCC for further processing and notification.



Environment Protection Act, 1986



- Western Ghats
- Gadgil panel (2011) recommended tagging 64% of the Western Ghats region as an Ecologically Sensitive Area (ESA) with varying degrees.
- Later Kasturirangan committee (2013) was appointed that proposed 37% total area of Western Ghats, to be declared as ESA.
 - Continued the existing system of 'red', 'orange' and 'green' categorisation of activities according to their polluting effects.
 - It recommended a blanket ban on mining, quarrying, setting up of red category industries and thermal power projects.

National Green Tribunal Act, 2010



- The Act provides for the establishment of a National Green Tribunal for the effective and expeditious disposal of cases relating to environmental protection
- NGT is a statutory body
 - Principal bench- Delhi
 - Regional benches in Pune (West), Bhopal (Central), Chennai (South) and Kolkata (East).

National Green Tribunal Act, 2010



- Composition
 - The Chairperson of the NGT is a retired Judge of the Supreme Court.
 - Other Judicial members are retired Judges of High Courts.
 - Each bench of the NGT will comprise at least one Judicial Member and one Expert Member.

National Green Tribunal Act, 2010



- Jurisdiction of NGT
 - The Tribunal is not bound by the procedure laid down under the Code of Civil Procedure 1908, but shall be guided by principles of 'natural justice'.
 - Sustainable development shall be the key guidance for any verdict of NGT
 - An appeal against verdict of the NGT lies to the Supreme Court, generally within ninety days.

National Green Tribunal Act, 2010



- NGT deals with the following Acts
 - The Water (Prevention and Control of Pollution) Act, 1974,
 - The Water (Prevention and Control of Pollution) Cess Act, 1977,
 - The Forest (Conservation) Act, 1980,
 - The Air (Prevention and Control of Pollution) Act, 1981,
 - The Environment (Protection) Act, 1986,
 - The Public Liability Insurance Act, 1991 and
 - The Biological Diversity Act, 2002.
- Note NGT **does not** deal with **Wildlife Protection Act, 1972** and Scheduled Tribes and Other Traditional Forest Dwellers (**Recognition of Forest Rights) Act, 2006**

Protection of Plant Varieties & Farmers' Rights (PPVFR) Act, 2001



- It has been enacted in India for giving effect to the TRIPS Agreement.
- The PPVFR Act retained the main spirit of TRIPS viz., IPRs as an incentive for technological innovation.
- However, the Act also had strong provisions to protect farmers' rights.
- The act allows farmers to plant, grow, exchange & sell patent-protected crops, including seeds, & only bars them from selling it as “branded seed”.
- It recognised three roles for the farmer: cultivator, breeder & conserver.
 - As cultivators, farmers were entitled to plant-back rights.
 - As breeders, farmers were held equivalent to plant breeders.
 - As conservers, farmers were entitled to rewards from a National Gene Fund.

PROTECTING PLANT VARIETIES



Technique for protecting plant varieties		
Technique	Benefits of the technique	Issues with the technique
Field gene bank	• Simple and traditional preservation strategy • Direct evaluation and characterisation • Seasonal accessibility and availability • Natural selection pressure	• Accumulation of mutations and endophytic organisms • Exposure of adverse weather conditions • Exposure to pathogens and insects • Plant ageing; Handling errors
In vitro gene bank	• Virus elimination • Immediate accessibility • Immediate availability • High multiplication rate • Low space requirements • Precise environment modulation • Medium term storage, < 2 years	• Plant ageing • Handling errors • Soma clonal variations • Specific protocol development • Infestation of insects (mites, thrips, other arthropods) • Contamination with fungi, bacteria & endophytic organism
Cryo bank	• Minimum space requirements • Low long-term costs • High genetic stability • Long-term storage, >100 years	• Restricted availability • Restricted accessibility • Access to liquid nitrogen • Specific protocol development • High initial workload to cryopreserve clonal plants

NATIONAL BOARD FOR WILDLIFE



- It is a Statutory Organization constituted under the WPA, 1972.
- It was constituted in 2003 replacing Indian Board For Wildlife (IBWL)
- Primary function of the Board is to promote the conservation and development of wildlife and forests.
- It serves as apex body to review all wildlife-related matters and approve projects in and around national parks and sanctuaries.
- No alteration of boundaries in National Parks and Wildlife Sanctuaries can be done without approval of the NBWL.
- It is a 47-members body chaired by the Prime Minister and its vice chairman is the Minister of Environment.

ANIMAL WELFARE BOARD OF INDIA (AWBI)



- Established in 1962 under the Prevention of Cruelty to Animals Act, 1960.
- Started under the stewardship of Late **Smt. Rukmini Devi Arundale**.
- Government has shifted headquarters of Animal Welfare Board of India (AWBI) to Ballabhgarh in Faridabad District of Haryana from Chennai, Tamil Nadu.
- The Board was initially within the jurisdiction of the GOI's Ministry of Food and Agriculture. And later in 1990, the subject of Prevention of Cruelty to Animals was transferred to the Ministry of Environment and Forests.

ANIMAL WELFARE BOARD OF INDIA (AWBI)



FUNCTIONS OF AWBI:

- ensures that animal welfare laws in the country are diligently followed, to provide grants to Animal Welfare Organizations
- suggests changes to laws and rules about animal welfare issues.
- issues publications to raise awareness of various animal welfare issues.
- Recognises Animal Welfare Organisations.

WILDLIFE INSTITUTE OF INDIA (WII)



- Established as an attached office in May, 1982.
- Granted autonomous status under the MoEFCC in 1985.
- Function: conducts specialised research in areas of study like Endangered Species, Biodiversity, Wildlife Management, Wildlife Policy, Wildlife Forensics, Habitat Ecology, Spatial Modelling, Eco-development, and Climate Change.
- WII has a research facility which includes Forensics, Remote Sensing and GIS, Laboratory, Herbarium, and an Electronic Library.

WILDLIFE INSTITUTE OF INDIA (WII)



भारतीय वन्यजीव संस्थान
Wildlife Institute of India

CENTRAL POLLUTION CONTROL BOARD (CPCB)



- It is a statutory organization that was constituted in September 1974 under the Water (Prevention and Control of Pollution) Act, 1974.
- CPCB was entrusted with the powers and functions under the **Air (Prevention and Control of Pollution) Act, 1981.**

CENTRAL POLLUTION CONTROL BOARD (CPCB)



Functions of the CPCB:

- To promote cleanliness of streams and wells in different areas of the States through prevention, control, and abatement of water pollution, and
- To improve the quality of air and to prevent, control or abate air pollution in the country.
- It also provides technical services to the Ministry of Environment and Forests regarding the provisions of the Environment (Protection) Act, 1986.

FOREST SURVEY OF INDIA (FSI)



- It is an organization under the MoEFCC.
- Responsible for assessment and monitoring of the forest resources of the country regularly.
- Established in 1981.
- Headquartered at Dehradun.
- ISFR- India State of Forest Report is a biennial publication of Forest Survey of India (FSI), an organization under the Ministry of Environment Forest & Climate Change, since 1987.

FOREST SURVEY OF INDIA (FSI)



- The National Biodiversity Authority (NBA) was established in 2003 by the Central Government to implement India's Biological Diversity Act (2002).
- It is a Statutory body.
- It performs a facilitative, regulatory and advisory function for the Government of India on the issue of Conservation, sustainable use of biological resources and fair equitable sharing of benefits of use.
- The Biological diversity Act (2002) mandates implementation of the provisions of the Act through decentralized system.

NATIONAL BIODIVERSITY AUTHORITY

(NBA)

Structure of Bio-diversity Act

Three tier system

NBA

- ✓ Requests for access by foreign individuals, institutions or companies.
- ✓ Transfer of results of research to any foreigner.

SBB 29

- ✓ Access by Indians for commercial purposes.
- ✓ The Indian industry – provide about the use of Biological resources.

BMC

37,769

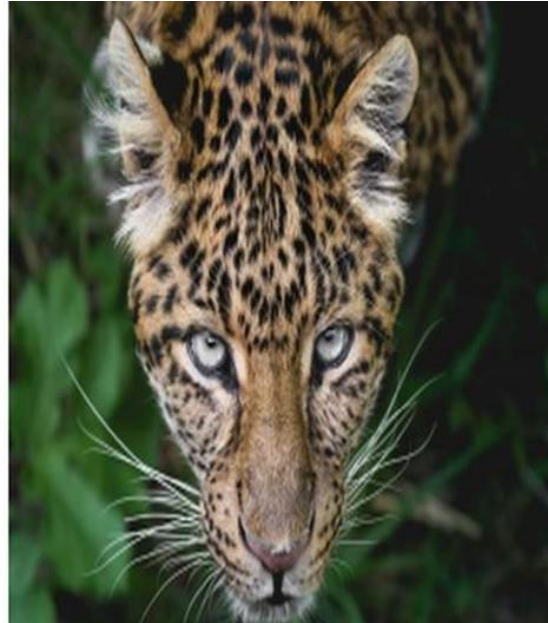
- ✓ Institutions of local self government.
- ✓ Conservation, sustainable use, documentation of biodiversity, etc.,

WILDLIFE CRIME CONTROL BUREAU (WCCB)



- WCCB is a statutory body established by the GOI under the MoEFCC, to combat organized wildlife crime in the country.
- It is the nodal agency in India for CITES related enforcements.
- It is also the nodal point for SAWEN in India
- United Nation Environment has awarded Wildlife Crime Control Bureau (WCCB), with Asia Environment Enforcement Awards, 2018 for excellent work done by the Bureau in combating transboundary environmental crime.

WILDLIFE CRIME CONTROL BUREAU (WCCB)



Mandate under Wild Life (Protection) Act, 1972:

- Collect and collate intelligence related to organized wildlife crime activities and to disseminate
- the same to State and other enforcement agencies;
- Establish a centralized wildlife crime data bank;
- Co-ordinate actions by various agencies in connection with the enforcement of the provisions of the Act;
- Assist foreign authorities and international organization concerned for wildlife crime control;

WILDLIFE CRIME CONTROL BUREAU (WCCB)



- Capacity building of the wildlife crime enforcement agencies for investigation into wildlife crimes and assist State Governments to ensure success in prosecutions related to wildlife crimes; and
- Advise the Government of India on issues relating to wildlife crimes

BOTANICAL SURVEY OF INDIA (BSI)



- It is the apex research organization under MoEFCC for carrying out taxonomic and floristic studies on wild plant resources of the country.
- It was established in 1890..
- Headquarters- Kolkata

Functions

- Publication of National, State and District Floras.
- Identification of threatened and red list species and species rich areas needing conservation.
- Ex-situ conservation of critically threatened species in botanical gardens.
- Survey and documentation of traditional knowledge (ethno-botany) associated with plants.
- Develop National database of Indian plants, including herbarium and live specimens, botanical paintings and illustrations, etc.

GENETIC ENGINEERING APPROVAL COMMITTEE



- GEAC is a statutory body constituted under the Environment (Protection) Act, 1986.
- It functions under the Ministry of Environment and Forests.
- The body regulates the use, manufacture, storage, import and export of hazardous
- microorganisms or genetically-engineered organisms and cells in India.

Functions

- It is the apex body to accord environmental approval of activities involving large scale use of
- hazardous microorganisms and recombinants in research and industrial production.
- It is also mandated with approving the release of genetically engineered organisms and
- products into the environment, including experimental field trials.
- The Committee has the power to take punitive action against people/body under the Environment (Protection) Act.

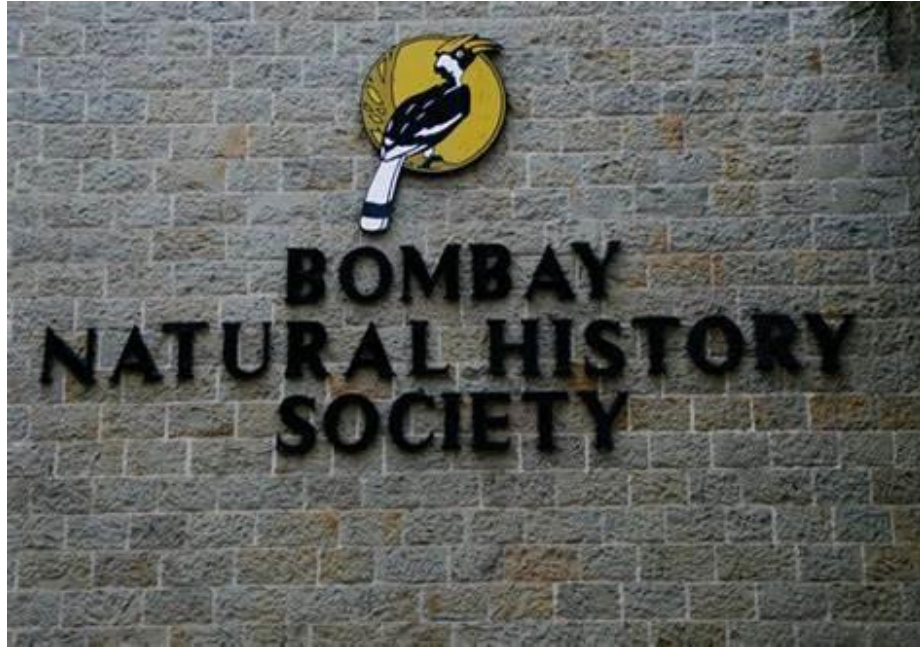
- The Central Zoo Authority of India (CZA) is the body of the government of India responsible for oversight of zoos.
- Headquarters: Delhi
- It is an affiliate member of the World Association of Zoos and Aquariums.
- It is a Statutory body, constituted under the Wild Life (Protection) Act.
- Central Zoo Authority is headed by **Minister of State for Environment & Forests (Forests & Wildlife)**, Government of India.

Functions

- Recognition and Evaluation of the Zoos
- Coordination in the planned conservation breeding programme for endangered species in Indian zoos
- Approval of the exchange proposals of animals between Indian zoos and between Indian and foreign zoos
- Preparing and maintaining the studbooks etc.

- BNHS is one of the largest non-governmental organisations in India engaged in conservation and biodiversity research.
- It was founded on 15 September 1883 and headquartered at Hornbill House, Mumbai.
- It supports many research efforts through grants and publishes the Journal of the Bombay Natural History Society.
- Department of Science and Technology has designated it as a 'Scientific and Industrial Research Organisation'.

Bombay Natural History Society (BHNS)



- Convention on Biological Diversity (CBD)
 - CBD is a legally binding treaty to conserve biodiversity that has been in force since 1993 and has been ratified by 196 nations.
 - It sets out guidelines for countries to protect biodiversity, ensure sustainable use, and promote fair and equitable benefit sharing.
 - India enacted Biological Diversity Act in 2002 for giving effect to the provisions of the CBD
 - CoP 15- Kunming-Montreal Global Biodiversity Framework
 - 4 goals and 23 targets for achievement by 2030.

International Measures



- Biodiversity Finance Initiative
 - UNDP
- High Ambition Coalition (HAC) for Nature and People
 - Central goal of protecting at least 30 percent of the world's land and ocean by 2030
 - India joined in 2021

Important Bird Areas (IBAs)



- Declared under IBA programme of Birdlife International
- The Bombay Natural History Society and Birdlife International identified 554 IBAs in India.

Aim:

- To identify, monitor and protect a global network of IBAs for conservation of the world's birds and associated biodiversity.
- Serve as conservation areas for protection of birds at the global, regional or sub-regional level.

Important Bird Areas (IBAs)



Criteria according to Birdlife International:

- hold significant numbers of one or more globally threatened bird species,
- be one of a set of sites that together hold a suite of restricted-range species or biome restricted species and
- have exceptionally large numbers of migratory or congregatory birds.

Natural World Heritage Sites



- Natural World Heritage Sites are listed by the United Nations Educational, Scientific and Cultural Organization as of special cultural or physical significance under World Heritage Convention of UNESCO.
- The World Heritage Convention is one of the most successful international instruments to protect the most extraordinary natural places on the planet, characterized by their natural beauty or outstanding biodiversity, ecosystem and geological values.

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- Was founded in October 1948 as the International Union for the Protection of Nature (or IUPN) following an international conference in Fontainebleau, France.
- Organization changed its name to the International Union for Conservation of Nature and Natural Resources in 1956 with the acronym IUCN (or UICN) with its headquarters in Gland, Switzerland.
- IUCN Members include both States and non-governmental organizations.

- Priority Areas
 - Biodiversity
 - Climate Change
 - Sustainable Energy
 - Human well-being
 - Green Economy

- Red Data Book
 - Red Data book or Red list is a catalogue of taxa facing risk of extinction.

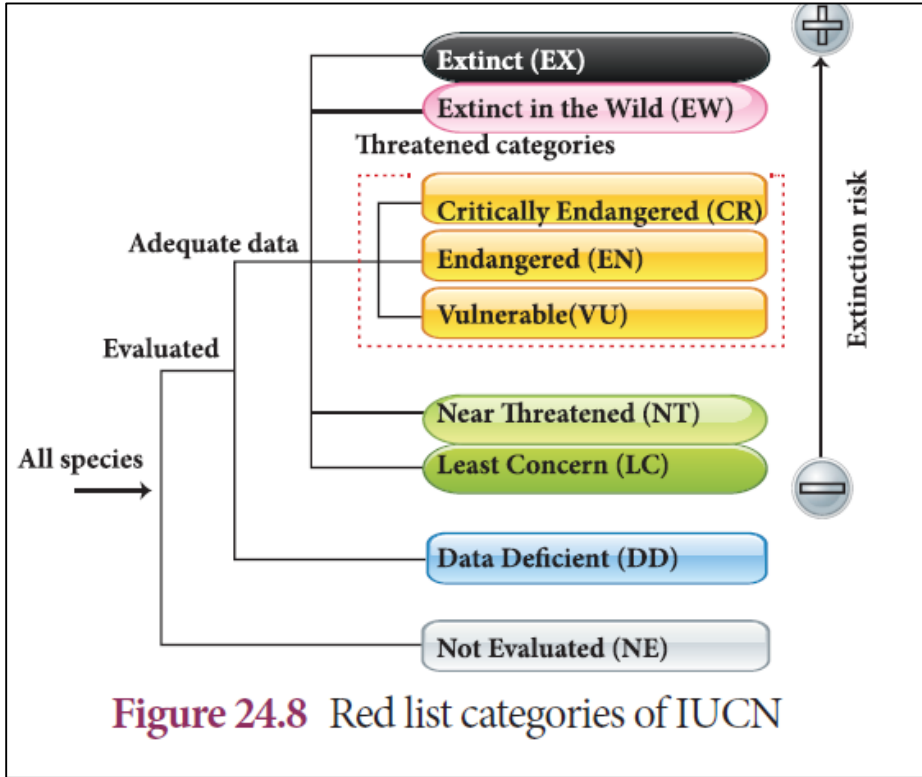


Figure 24.8 Red list categories of IUCN